

Fantasy Pathfinder

User Guide

Search, compare, and manage your fantasy football roster

Overview

Fantasy Pathfinder is a Streamlit app for exploring NFL player stats and managing a Sleeper fantasy league. It has two main areas, switched with the toggle at the top of the page:

- **Stat Search** - look up a player's stats, splits, and game log, or toggle on Compare Players to put up to 3 players side by side.
- **Sleeper** - connect your Sleeper account to view league rosters, get draft rankings, and find mutually beneficial trades.

Stat Search

This is the default view when you open the app.

1. Choose seasons

The Seasons field starts empty - pick one or more years before searching. If you search with nothing selected, the app falls back to the most recent seasons automatically.

2. Search for a player

Type a player's name into the search box. Suggestions appear as you type - pick one from the dropdown.

3. Optional: apply a split

The Split dropdown filters the game log to just the games matching that condition:

- **Location:** Home or Away
- **Roof:** Indoors or Outdoors
- **Day:** Weekday, Weekend, Monday, Short Week (Tuesday-Friday), Saturday, or Sunday
- **Time of day:** Europe, Early, Midday, or Night kickoff
- **Weather:** Rain, Snow, or Below 0 degrees
- **Divisional rivals only:** games against a division opponent

TIP: The split only filters the Game Log table and its chart. The Stat Splits table below always shows the full multi-way breakdown across every game, so it isn't affected by this dropdown.

4. Read the results

After clicking Search, the page shows:

- **Summary & Projection** - a quick projection with recent average, sample average, and trend direction.
- **Game Log** - a week-by-week table (Sleeper standard PPR scoring) with a bar chart of weekly fantasy points plus a 3-game rolling average line, and a snap % usage trend chart.
- **Fantasy Points By Category** - a bar chart showing how a player's fantasy points split across Passing, Rushing, and Receiving (only shown for players who score in 2 or more categories).
- **Stat Splits** - an ESPN-style table breaking the season down by Home/Away, Outcome, Victory Margin, Month, Day, Surface, Location, Weather, and Group (division/conference), with a chart picker (Chart Split) to visualize any one of those breakdowns.

Compare Players

Flip the Compare Players toggle (just below the search box) to compare players instead of searching one. Your original search box becomes Player 1, and two more search boxes appear for Player 2 and Player 3 - fill in at least one to compare.

- **Split filter** - applies the same way as in Stat Search, restricting the comparison to games matching that split.
- **Comparison table** - season totals/averages plus position-appropriate stat groups (Passing, Rushing, Receiving), with the best value in each row highlighted.
- **Chart Stat** - a dropdown under the table lets you pick any single stat (Season Total, Season Avg, or any group stat like Rushing YDS or Receiving TD) to chart across the compared players.

TIP: Turn the toggle back off to return to a single-player search - your Player 1 entry carries back over as the search box.

Sleeper Integration

Switch to the Sleeper section to connect your fantasy league.

1. Find your leagues

Enter your Sleeper username and pick a season, then click Find My Leagues.

2. View rosters

Once leagues are found, pick one from Select A League and click View Rosters. Your team appears first under Your Team, followed by every other team in the league as collapsible sections.

3. Draft Optimizer

After your league loads, a Draft Optimizer button appears. It ranks available players by projected value over a replacement-level player at the same position:

- **Players Already Drafted** - remove players who are already off the board.
- **Rank By Position** - filter the rankings to one position, or view All Positions.
- **Column Guide** - an expandable reference explaining Predicted Points, Replacement Points, and Value Over Replacement.

TIP: The optimizer currently uses sample projections until the full prediction model output is connected.

4. Trade Finder

Also appears after your league loads. Set a Minimum weekly gain for both teams (the smallest projected improvement either side must see for a trade to qualify), then click Find Trade Ideas. Results show, for each proposed trade: your team and the trade partner, what each side gives up, and the projected weekly point gain for each team plus the combined gain.

Tips & Troubleshooting

- **No results after searching?** make sure at least one season is selected and a player is chosen from the autocomplete dropdown, not just typed.
- **Compare warning about 2 players?** the search box at the top counts as Player 1 - fill in at least one of the two boxes below it too.
- **Draft Optimizer / Trade Finder missing?** both only appear in the Sleeper section after you've found your leagues (Draft Optimizer) or after rosters are loaded (Trade Finder).
- **Trade Finder found nothing?** try lowering the Minimum weekly gain value.